

LAB 1

The screenshot shows two side-by-side browser windows. The left window displays the Google Skills interface for the task "Grant an IAM role". It includes a sidebar with navigation options like Dashboard, Catalog, Paths, Collections, and Subscriptions. The main content area provides instructions for granting an IAM role to a principal. A progress indicator in the top right corner shows "0/100".

The right window shows the Google Cloud IAM & Admin console for the project "qwklabs-gcp-00-b2e5afa3e67d". The "IAM" page is active, showing a list of principals. The "Grant access" button is visible. A table lists the principals and their roles:

Principal	Name	Role
admiral@qwklabs-services-prod.iam.gserviceaccount.com		Owner
qwklabs-gcp-00-b2e5afa3e67d@qwklabs-gcp-00-b2e5afa3e67d.iam.gserviceaccount.com	Qwiklabs User Service Account	BigQuery Admin, Owner, Storage Admin
student-04-669f3973814a@qwklabs.net		Viewer
student-04-d4d577b5b6f5@qwklabs.net	student cab0c9b43	Editor, Project IAM Ac, Viewer

A notification at the bottom of the right window states: "Policy updated. It may take a few minutes for these changes to become active." with an "OK, got it" button.

The screenshot shows the Google Cloud product details page for the Dialogflow API. The page includes the Dialogflow logo, a description "Builds conversational interfaces", and a "Manage" button. The "API Enabled" status is shown with a green checkmark. The "Overview" section provides details about the API, including its type (SaaS & APIs), last product update (4/30/22), category (Machine learning, Google Enterprise APIs), and service name (dialogflow.googleapis.com). The "Pricing" section is also visible. A notification at the bottom of the page states: "Policy updated. It may take a few minutes for these changes to become active." with an "OK, got it" button.

Enabled API and granted an IAM role.

Created a vm instance

The screenshot shows the Google Cloud console interface. On the left is a navigation menu with options like Overview, Security risk overview, Virtual machines, VM instances, Instance templates, Sole-tenant nodes, Machine images, TPUs, Committed use discounts, Reservations, Migrate to Virtual Machines, Storage, Disks, Storage Pools, Snapshots, Images, Async Replication, Instance groups, Instance groups, Health checks, Marketplace, and Release Notes. The main area is titled 'VM instances' and includes a 'Create Instance' button. Below this is a table of VM instances:

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
Running	gcelab	us-east5-a			10.202.0.2 (nic0)	34.162.113.204 (nic0)	SSH

Below the table are 'Related actions' such as 'Fix data protection gaps', 'Monitor VMs', 'Explore VM logs', 'Set up firewall rules', and 'Load balance between VMs'. On the right, there is a 'Create a Virtual Machine' tutorial for GSP001.

@Installed nginx web server

The screenshot shows a terminal window with the following output:

```
student-01-76b225f3e4c2@gcelab:~$ sudo apt-get install -y nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  logrotate nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 641 kB of archives.
After this operation, 1696 kB of additional disk space will be used.
Get:1 file:/etc/apt/sources/debian-security.list Mirrorlist [39 B]
Get:2 https://deb.debian.org/debian-security bookworm-security/main amd64 nginx-common all 1.22.1-9+deb12u4 [3 KB]
Get:3 https://deb.debian.org/debian-security bookworm-security/main amd64 nginx amd64 1.22.1-9+deb12u4 [520 kB]
Fetched 641 kB in 0s (4259 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 76012 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.22.1-9+deb12u4_all.deb ...
Unpacking nginx-common (1.22.1-9+deb12u4) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.22.1-9+deb12u4_amd64.deb ...
Unpacking nginx (1.22.1-9+deb12u4) ...
Setting up nginx-common (1.22.1-9+deb12u4) ...
Use of the nginx package requires the presence of the nginx.service file.
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /lib/systemd/system/nginx.service.
Setting up nginx (1.22.1-9+deb12u4) ...
Upgrading binary: nginx.
Processing triggers for man-db (2.11.2-2) ...
student-01-76b225f3e4c2@gcelab:~$ ps aux | grep nginx
root      1365  0.0  0.1 10376  6656 ?        Ss   22:18   0:00 nginx: master process /usr/sbin/nginx -g d
non-ou/ master process on
www-data  1367  0.0  0.2 25576  9448 ?        Ss   22:18   0:00 nginx: worker process
www-data  1368  0.0  0.2 25576  9456 ?        Ss   22:18   0:00 nginx: worker process
student-01-76b225f3e4c2@gcelab:~$
```

Part 2, Creating a virtual machine

@Created a vm instance using gcloud

The screenshot shows a Google Skills lab interface on the left and a Cloud Shell terminal on the right. The lab interface displays the command `gcloud compute instances create gcelab2 --machine-type e2-medium --zone=$ZONE` and its expected output. The terminal shows the execution of this command, which results in an error: `ERROR: (gcloud.compute.instances.create) could not parse resource []`. The error message indicates that the `--zone` flag is required. The lab interface also shows a green checkmark indicating that the assessment is completed.

Expected output:

```
Created [...gcelab2].
NAME: gcelab2
ZONE: us-east5-a
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.128.0.3
EXTERNAL_IP: 34.136.51.150
STATUS: RUNNING
```

To check your progress in this lab, click **Check my progress** below. A checkmark means you're successful.

Create a new instance with gcloud.

Assessment Completed!

The new instance has these default values:

- The [Debian GNU/Linux 12 \(bookworm\)](#) image.
- The [e2-medium machine type](#).
- A root persistent disk with the same name as the instance; the disk is automatically attached to the instance.

Cloud Shell terminal output:

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-01-916c4634bfcd.
Use "gcloud config set project [PROJECT_ID]" to change to a different project.
student_01_76b225f3e4c2@cloudshell:~$ gcloud config set compute/region us-east5
Updated property [compute/region].
student_01_76b225f3e4c2@cloudshell:~$ (quiklabs-gcp-01-916c4634bfcd) $ gcloud config set compute/region us-east5
student_01_76b225f3e4c2@cloudshell:~$ (quiklabs-gcp-01-916c4634bfcd) $ gcloud config set compute/region us-east5
student_01_76b225f3e4c2@cloudshell:~$ (quiklabs-gcp-01-916c4634bfcd) $ gcloud compute instances create gcelab2 --ma
e2-medium --zone=$ZONE
ERROR: (gcloud.compute.instances.create) could not parse resource []
student_01_76b225f3e4c2@cloudshell:~$ (quiklabs-gcp-01-916c4634bfcd) $ ^C
student_01_76b225f3e4c2@cloudshell:~$ (quiklabs-gcp-01-916c4634bfcd) $ gcloud compute instances create gcelab2 --ma
e2-medium --zone=$ZONE
ERROR: (gcloud.compute.instances.create) Could not fetch resource:
- Permission denied on 'locations/zone' (or it may not exist).
student_01_76b225f3e4c2@cloudshell:~$ (quiklabs-gcp-01-916c4634bfcd) $ gcloud compute instances create gcelab2 --ma
e2-medium --zone=$ZONE
Created [https://www.googleapis.com/compute/v1/projects/quiklabs-gcp-01-916c4634bfcd/zones/us-east5-a/instances/g
NAME: gcelab2
ZONE: us-east5-a
MACHINE_TYPE: e2-medium
PREEMPTIBLE:
INTERNAL_IP: 10.202.0.3
EXTERNAL_IP: 34.162.174.97
STATUS: RUNNING
student_01_76b225f3e4c2@cloudshell:~$ (quiklabs-gcp-01-916c4634bfcd) $
```

The screenshot shows a Google Skills lab interface on the left and a Google Cloud console on the right. The lab interface displays the commands `gcloud config set compute/zone ...` and `gcloud config set compute/region ...`, and a list of steps for creating a VM instance. The Google Cloud console shows the 'VM instances' page with a table listing the instances 'gcelab' and 'gcelab2'. A 'Chat with Gemini Cloud Assist' window is open, providing a tip to use Alt+G to open and close the chat.

To do this, run these commands:

```
gcloud config set compute/zone ...
gcloud config set compute/region ...
```

100/100

3. To exit `help`, press **CTRL+C**.

4. In the Cloud console, on the **Navigation menu** (☰), click **Compute Engine > VM instances**. Or if you still had the VM instances page open, just click **Refresh**. Your two new instances should be listed.

5. You can also use SSH to connect to your instance via `gcloud compute ssh`. Make sure to add your zone, or omit the `--zone` flag and set the option globally:

```
gcloud compute ssh gcelab2 --zone=us-east5-a
```

6. Enter **Y** to continue.

Do you want to continue? (Y/n)

7. Press **Enter** through the passphrase section to leave passphrase empty.

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase)

Google Cloud console 'VM instances' page:

Status	Name	Zone	Connect
☒	gcelab	us-	SSH
☒	gcelab2	us-	SSH

7. Press Enter under the passphrase section to leave the passphrase empty.

8. After connecting, disconnect from SSH by exiting from remote shell with the command that follows:

```
exit
```


The screenshot shows the Google Cloud Shell Editor interface. The terminal window displays the output of the 'exit' command, which ends the SSH session. The output includes the key's random image, the warning about permanently added hosts, and the programs included with the Debian GNU/Linux system.

Created a windows server virtual machine

The image is a composite of two parts: a document page on the left and a Google Cloud console screenshot on the right.

Document Page (Left):

- Header:** Google Skills
- Navigation:** Dashboard, Catalog, Paths, Collections, Subscriptions.
- Content:**
 - Click **Select**.
 - 4. Click **Create** to create the instance.
 - Test completed task**
 - Click **Check my progress** to verify your performed task.
 - Create a virtual machine instance.
 - [Check my progress](#)
 - Remote Desktop (RDP) into the Windows Server**
 - Test the status of Windows Startup**
 - After a short time, the Windows Server instance will be provisioned and listed on the VM Instances page with a green status icon.
 - The server instance may not be ready to accept RDP connections, as it takes a while for all OS components to initialize.
 - 1. To see whether the server instance is ready for an RDP connection, run the following command at your Cloud Shell terminal command line and please make to replace [instance] with the VM Instance that you created earlier.

Google Cloud Console Screenshot (Right):

- Header:** Google Cloud
- Navigation:** Overview, Security risk overview, Virtual machines, VM instances, Instance templates, Sole-tenant nodes, Machine images, TPUs, Committed use discounts, Reservations, Migrate to Virtual Machines, Storage, Disks, Storage Pools, Snapshots, Images, Async Replication, Instance groups, Instance groups, Health checks, Marketplace, Release Notes.
- Content:**
 - Compute Engine**
 - VM instances**
 - [Create instance](#) [Import VM](#)
 - [Instances](#) [Observability](#) [Instance schedules](#)
 - VM instances**
 - [Filter](#) Enter property name or value
 - | Status | Name ↑ | Zon | Connect |
|-------------------------------------|---------------|------|---------|
| ☒ | windowsserver | US-1 | RDP |
 - Related actions** [Hide](#)

Windowsserver is ready to use and we can now connect using RDP

```
2026-03-26T22:37:07.3779Z CorePlugin: [INFO]: Setting up Windows accounts.
2026-03-26T22:37:07.3779Z CorePlugin: [INFO]: No new keys found, skipping account setup.
2026-03-26T22:37:07.6571Z CorePlugin: [INFO]: Instance id is already set, no update required.
2026-03-26T22:37:08.0536Z CorePlugin: [INFO]: Instance has service account: true, setting ACS client isEnabled to true
2026-03-26T22:37:09.0299Z CorePlugin: [INFO]: Skipping schedule job request for "gce-workload-cert-refresher"
2026-03-26T22:37:09.0305Z CorePlugin: [INFO]: Creating metric registry for "events_manager" with flush interval: 1m0s, max
records: 10
2026-03-26T22:37:12.7431Z CorePlugin: [INFO]: Cloud logging initialized (CorePlugin.exe)
2026/03/26 22:37:31 GCEInstanceSetup: Activating Windows(R), ServerDatacenter edition (ef6cfc9f-8c5d-44ac-9aad-de6a2ea0ae03)
...
2026/03/26 22:37:31 GCEInstanceSetup: Product activated successfully.
2026/03/26 22:37:32 GCEInstanceSetup: Activation successful.
2026/03/26 22:37:32 GCEInstanceSetup: Running 'schtasks' with arguments '/change /tn GCESStartup /enable'
2026/03/26 22:37:33 GCEInstanceSetup: --> SUCCESS: The parameters of scheduled task "GCESStartup" have been changed.
2026/03/26 22:37:33 GCEInstanceSetup: Running 'schtasks' with arguments '/run /tn GCESStartup'
2026/03/26 22:37:33 GCEInstanceSetup: --> SUCCESS: Attempted to run the scheduled task "GCESStartup".
2026/03/26 22:37:33 GCEInstanceSetup: -----
2026/03/26 22:37:33 GCEInstanceSetup: Instance setup finished. windowsserver is ready to use.
2026/03/26 22:37:33 GCEInstanceSetup: -----
2026-03-26T22:37:34.3244Z [INFO]: Initialized Metadata Script Runner Compat (version 20260228.00)
2026-03-26T22:37:34.3654Z [INFO]: Enable core plugin set to: [true], launching script runner for event "startup" from "C:\P
rogram Files\Google\Compute Engine\agent\GCEMetadataScriptRunner.exe"
2026-03-26T22:37:34.9234Z [INFO]: Running metadata script runner (version 20260228.00) for "startup" event
2026-03-26T22:37:35.0391Z [INFO]: No startup scripts to run
2026-03-26T22:37:55.5506Z OSConfigAgent Info: Enforce state: resource "install-pkg" enforcement successful.
2026-03-26T22:37:55.7420Z OSConfigAgent Info: Check state post enforcement: resource "add-repo" state is COMPLIANT.
2026-03-26T22:37:55.9085Z OSConfigAgent Info: Policy "goog-ops-agent-policy" resource "add-repo" state: COMPLIANT
2026-03-26T22:37:57.0107Z OSConfigAgent Info: Check state post enforcement: resource "install-pkg" state is COMPLIANT.
2026-03-26T22:37:57.0627Z OSConfigAgent Info: Policy "goog-ops-agent-policy" resource "install-pkg" state: COMPLIANT
2026-03-26T22:37:57.1902Z OSConfigAgent Info: Successfully completed ApplyConfigTask
```

Specify --start=24293 in the next get-serial-port-output invocation to get only the new output starting from here.
student_04_02148842adaf@cloudshell:~ (qwiklabs-gcp-02-3e4be114f9c2) \$ gcloud compute instances get-serial-port-output --start
=24293 windowsserver --zone=us-east4-b

Specify --start=24293 in the next get-serial-port-output invocation to get only the new output starting from here.
student_04_02148842adaf@cloudshell:~ (qwiklabs-gcp-02-3e4be114f9c2) \$

The screenshot shows a Google Cloud console interface with a sidebar on the left containing navigation links like 'Google Skills', 'Dashboard', 'Catalog', 'Paths', 'Collections', and 'Subscriptions'. The main content area displays 'Congratulations!' and 'Next steps/Learn more' sections. Overlaid on this is a remote desktop window titled 'windowsserver - 136.107.230.12 - Remote Desktop Connection'. The remote desktop shows a Windows Server desktop with icons for 'Recycle Bin', 'This PC', 'Google Cloud', 'SPK Shell', and 'Microsoft Edge'. In the background, the Google Cloud console shows the 'VM instances' page for the project 'qwiklabs-gcp-02-3e4be114f9c2'. A table lists the VM instances:

Instances	In use by	Internal IP	External IP	Connect
		10.150.0.2 (nic0)	136.107.230.12 (nic0)	RDP

The bottom of the screenshot shows a Windows taskbar with a search bar, task icons, and system tray information indicating the time is 6:48 PM on 3/26/2026.

Part 4, cloud shell and gcloud

Google cloud command line

The screenshot shows a Google Cloud documentation page on the left and a Cloud Shell terminal on the right. The documentation page has a sidebar with navigation links: Dashboard, Catalog, Paths, Collections, and Subscriptions. The main content area is titled "Finding project information" and contains three numbered steps:

- Copy your project ID to your clipboard or text editor. The project ID is in 2 places:
 - In the Cloud Console, on the Dashboard, under **Project info**. (Click the **Navigation menu** (☰), and then click **Cloud overview** > **Dashboard**.)
 - On the lab tab near your username and password.
- In Cloud Shell, run the following `gcloud` command, to view the project name of your project:

```
gcloud config get-value project
```
- In Cloud Shell, run the following `gcloud` command to view details about your project:

The Cloud Shell terminal on the right shows the following commands and output:

```
gcloud config set compute/zone asia-southeast1-a
gcloud config get-value compute/zone
gcloud config get-value project
gcloud compute project-info describe --project $(gcloud config get-value project)
```

The terminal output shows the project details, including the project name, region, and zone.

Details about the project

The screenshot shows a Google Cloud documentation page on the left and a Cloud Shell terminal on the right. The documentation page has a sidebar with navigation links: Dashboard, Catalog, Paths, Collections, and Subscriptions. The main content area is titled "Details about the project" and contains three numbered steps:

- In Cloud Shell, run the following `gcloud` command, to view the project name of your project:

```
gcloud config get-value project
```
- In Cloud Shell, run the following `gcloud` command to view details about the project:

```
gcloud compute project-info describe --project $(gcloud config get-value project)
```
- Find the zone and region metadata values in the output. You'll use the zone (google-compute-default-zone) from the output later in this lab.

The Cloud Shell terminal on the right shows the following commands and output:

```
gcloud config get-value project
gcloud compute project-info describe --project $(gcloud config get-value project)
```

The terminal output shows the project details, including the project name, region, and zone.

```
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $ echo -e "PROJECT ID: $PROJECT_ID\nZONE: $ZONE"
PROJECT ID: qwiklabs-gcp-04-445dc331e610
ZONE: asia-southeast1-a
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $
```

```
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610)$ gcloud compute instances create gcelab2 --machine-type
e2-medium --zone $ZONE
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-445dc331e610/zones/asia-southeast1-a/instances/gcela
b2].
NAME: gcelab2
ZONE: asia-southeast1-a
MACHINE TYPE: e2-medium
PREEMPTIBLE:
INTERNAL IP: 10.148.0.2
EXTERNAL IP: 136.110.44.33
STATUS: RUNNING
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610)$
```

Some gcloud commands

```
Cloud Shell Editor
qwiklabs-gcp-04-445dc331e610
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more
Don't show again Dismiss
To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610)$ gcloud compute firewall-rules list --filter="NETWORK:'default' AND ALLOW:'icmp'"
NAME: default-allow-icmp
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:0-65535,udp:0-65535,icmp
DENY:
DISABLED: False
NAME: default-allow-internal
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:1389
DENY:
DISABLED: False
NAME: default-allow-ssh
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:22
DENY:
DISABLED: False
To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610)$
```

Connected

```
WARNING: The private SSH key file for gcloud does not exist.
WARNING: The public SSH key file for gcloud does not exist.
WARNING: You do not have an SSH key for gcloud.
WARNING: SSH keygen will be executed to generate a key.
This tool needs to create the directory [/home/student_04_14c8443d0f29/.ssh] before being able to
generate SSH keys.

Do you want to continue (Y/n)? y

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/student_04_14c8443d0f29/.ssh/google_compute_engine
Your public key has been saved in /home/student_04_14c8443d0f29/.ssh/google_compute_engine.pub
The key fingerprint is:
SHA256:xlDlCkwySo4IDa3ib2c+PU3jNRLGpmd68+FDkVf6U8 student_04_14c8443d0f29@cs-730207478656-default
The key's randomart image is:
+----[RSA 3072]-----+
|      .      .      |
|B=O . O .      |
|+oo B O . o |
|o . = * + . o |
|o . S . . . .E|
| . O + . O . .|
| . O + + . . |
| O + + . O . |
| . + . . O O O + |
+----[SHA256]-----+
Warning: Permanently added 'compute.3625459752502821959' (ED25519) to the list of known hosts.
Linux gcelab2 6.1.0-43-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.162-1 (2026-02-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-04-14c8443d0f29'.
student-04-14c8443d0f29@gcelab2:~$
```

Added the tag and updated firewall rules

```
Cloud Shell Editor

(qwiklabs-gcp-04-445dc331e610) x + v x

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

DISABLED: False
NAME: default-allow-rdp
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:3389
DENY:
DISABLED: False
NAME: default-allow-ssh
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 65534
ALLOW: tcp:22
DENY:
DISABLED: False

To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.

student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $ gcloud compute instances add-tags gcelab2 --tags http-server,https-server
Updated [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-445dc331e610/zones/asia-southeast1-a/instances/gcelab2].
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $ gcloud compute firewall-rules create default-allow-http --direction=INGRESS --priority=1000 --network=default --action=ALLOW --rules=tcp:80 --source-ranges=0.0.0.0/0 --target-tags=http-server
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-445dc331e610/global/firewalls/default-allow-http].
Creating firewall...done.
NAME: default-allow-http
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:80
DENY:
DISABLED: False
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $
```

Listed the firewall rules and verified communication is possible for http to the virtual machine

```
Cloud Shell Editor

(qwiklabs-gcp-04-445dc331e610) x + v x

Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more Don't show again Dismiss

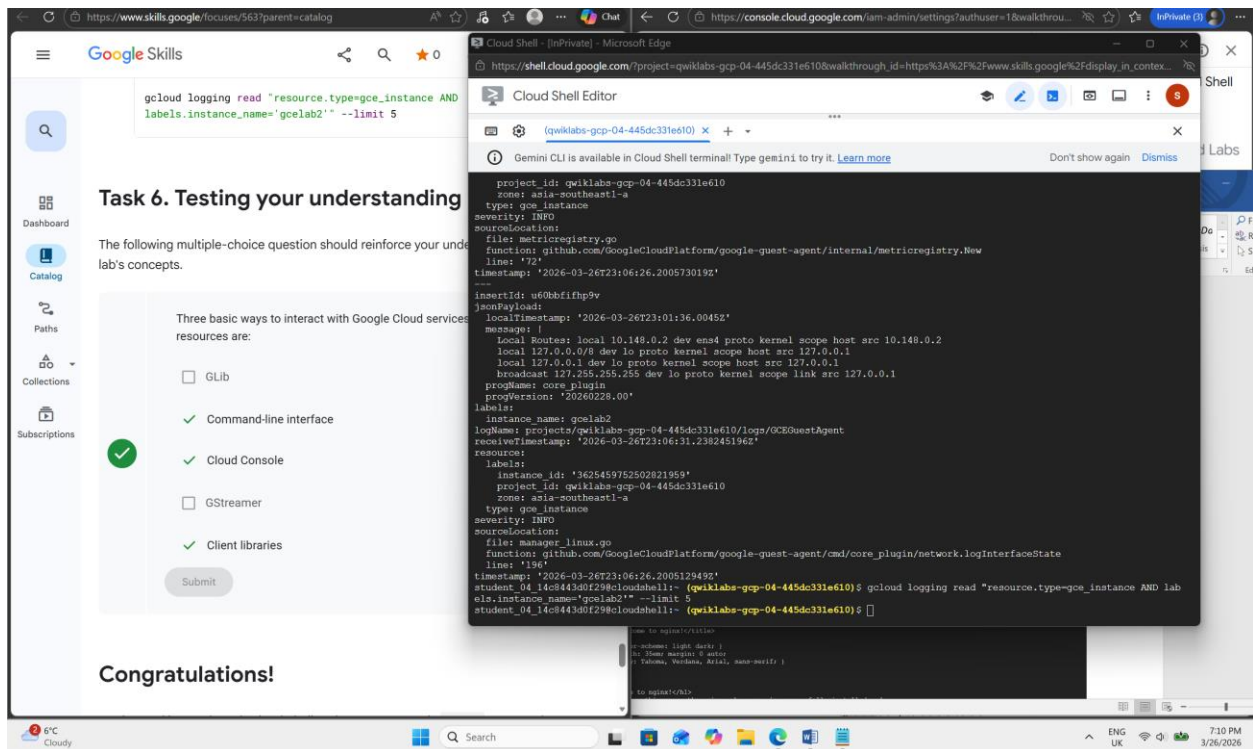
DISABLED: False
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $ gcloud compute firewall-rules list --filter=ALLOW:'80'
NAME: default-allow-http
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:80
DENY:
DISABLED: False

To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.

student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $ curl http://$(gcloud compute instances list --filter=name:gcelab2 --format='value(EXTERNAL_IP)')
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color:scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

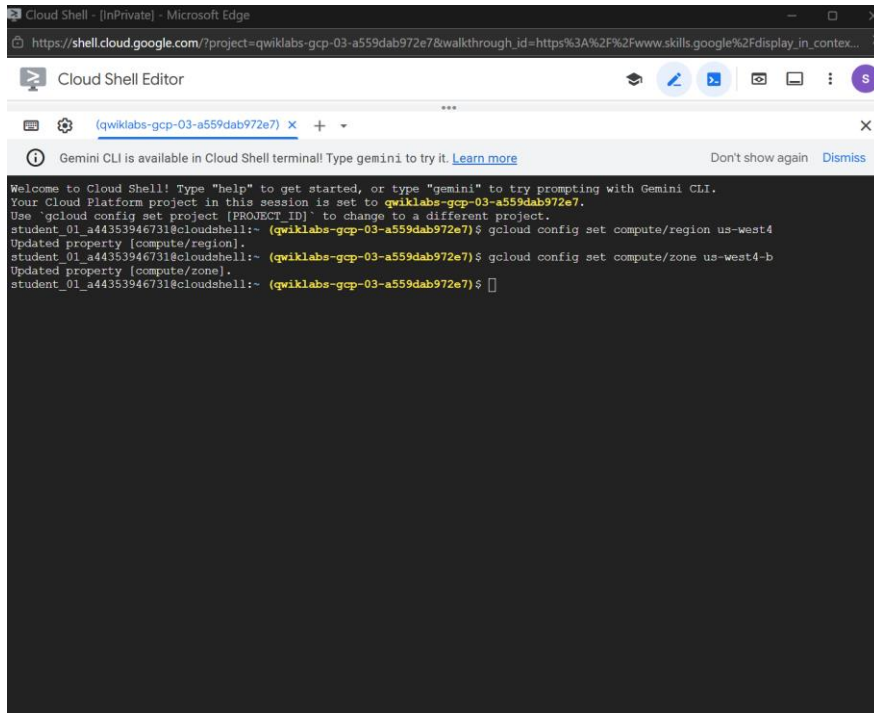
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
student_04_14c8443d0f29@cloudshell:~ (qwiklabs-gcp-04-445dc331e610) $
```

@system logs and viewer

PART 5 - Kubernetes Engine: Qwik Start

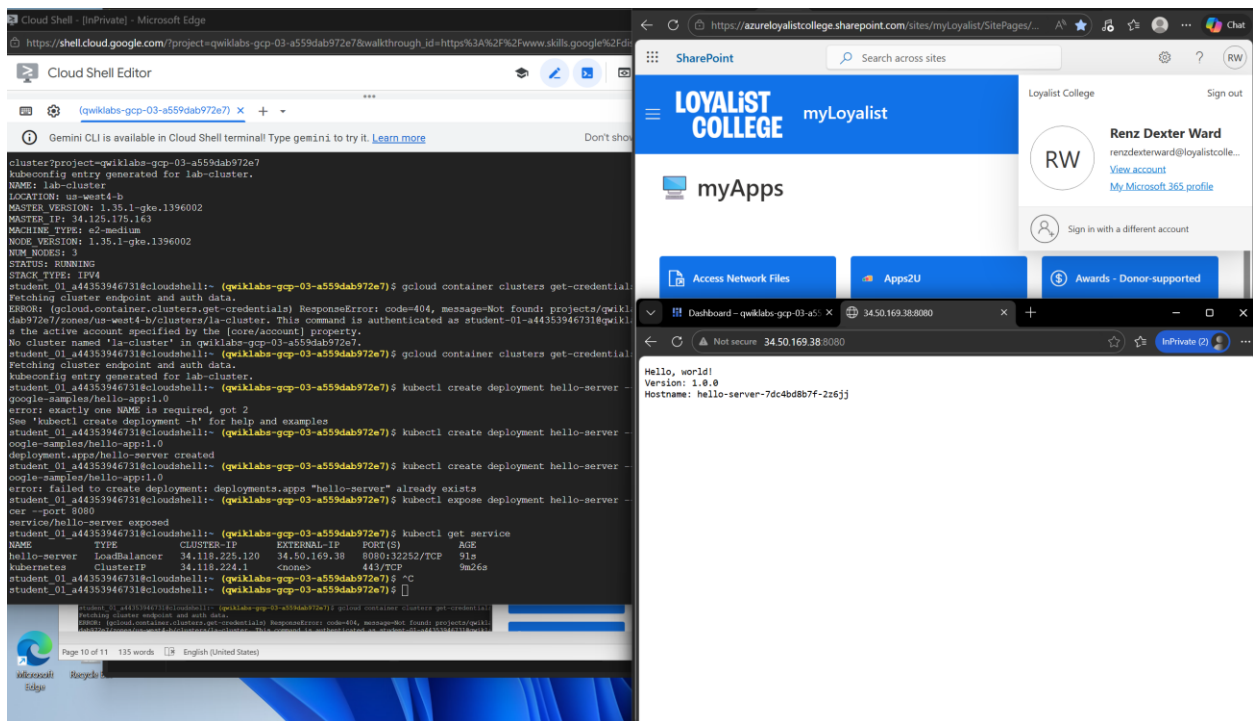
Set the default compute region and zone



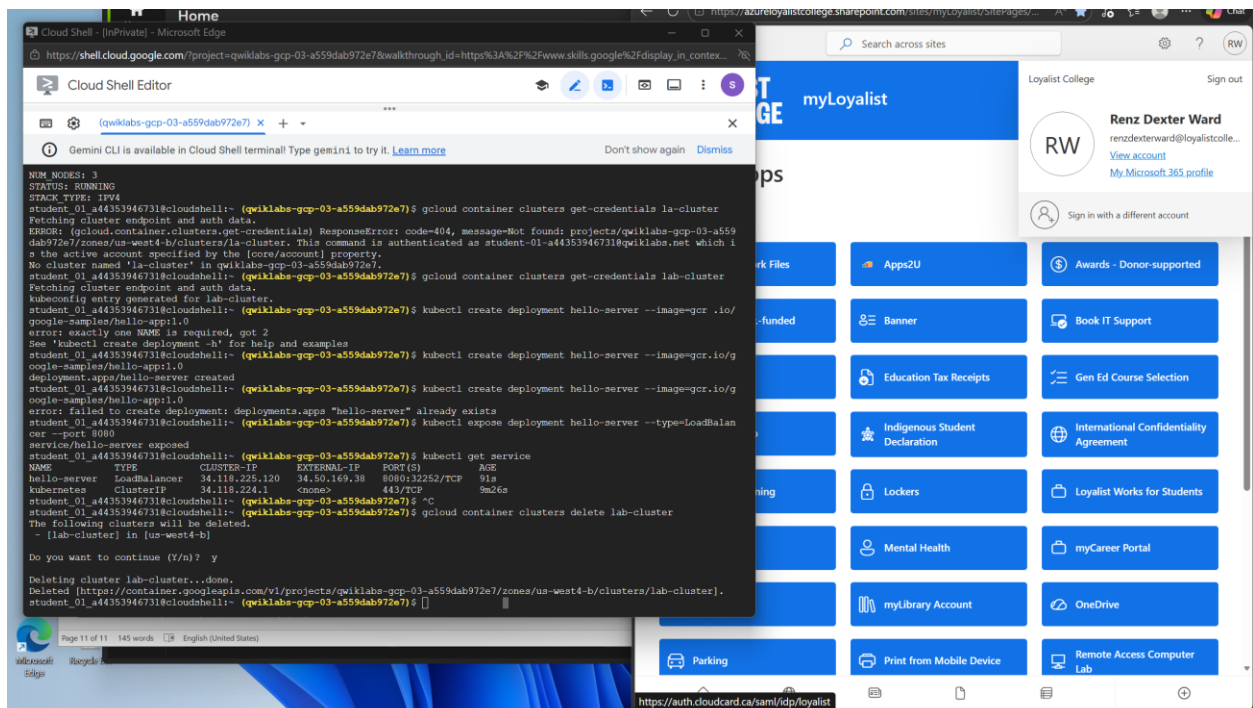
The screenshot displays two overlapping browser windows. The background window is Google Cloud Shell, showing a terminal session where a user creates a new Kubernetes cluster named 'lab-cluster'. The terminal output includes project details, configuration steps, and successful creation messages. A green 'GET' button is visible at the bottom of the terminal window.

The foreground window shows the 'myLoyalist' website. It features a blue header with the site name and navigation links like 'Home', 'About Us', 'Contact Us', and 'FAQs'. Below the header is a grid of service tiles including 'Apps2U', 'Banner', 'Education Tax Receipts', 'Indigenous Student Declaration', 'Lockers', 'Mental Health', 'myLibrary Account', 'Parking', 'Print from Mobile Device', 'Awards - Donor-supported', 'Book IT Support', 'Gen Ed Course Selection', 'International Confidentiality Agreement', 'Loyalist Works for Students', 'myCareer Portal', 'OneDrive', and 'Remote Access Computer Lab'. On the right side, there's a profile section for 'Renz Dexter Ward' with a login link and social media profiles.

Inspected the server and viewed the application from a browser

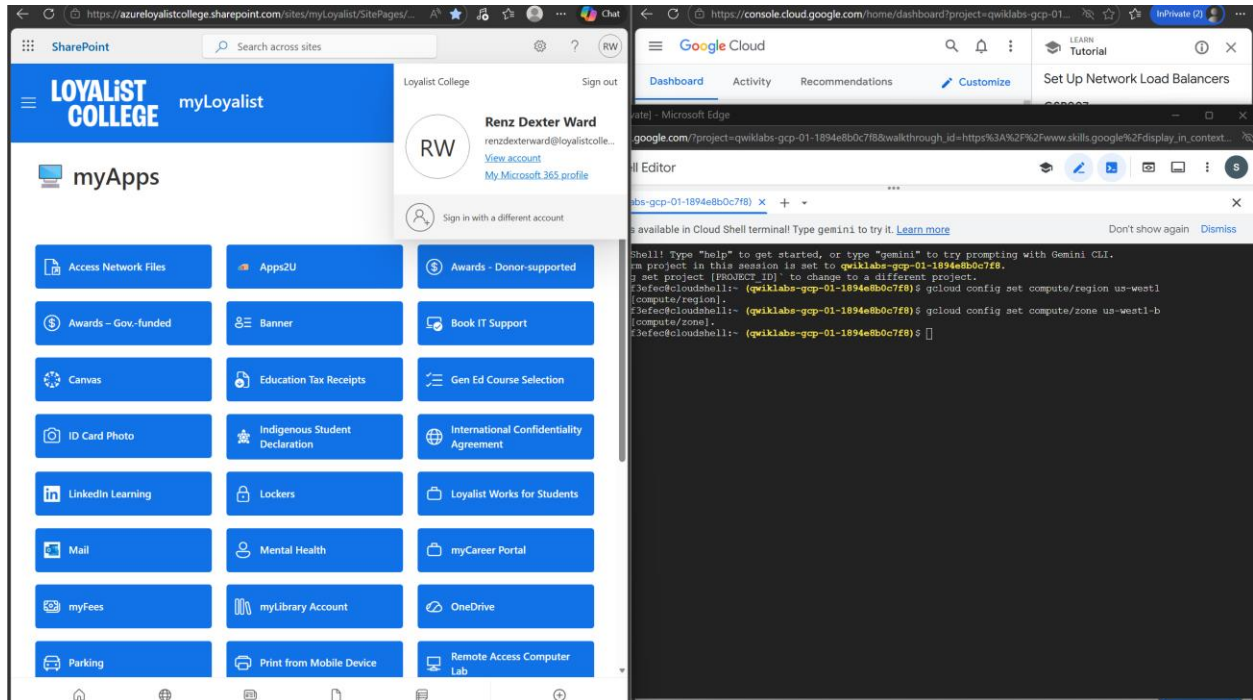


Deleted the cluster

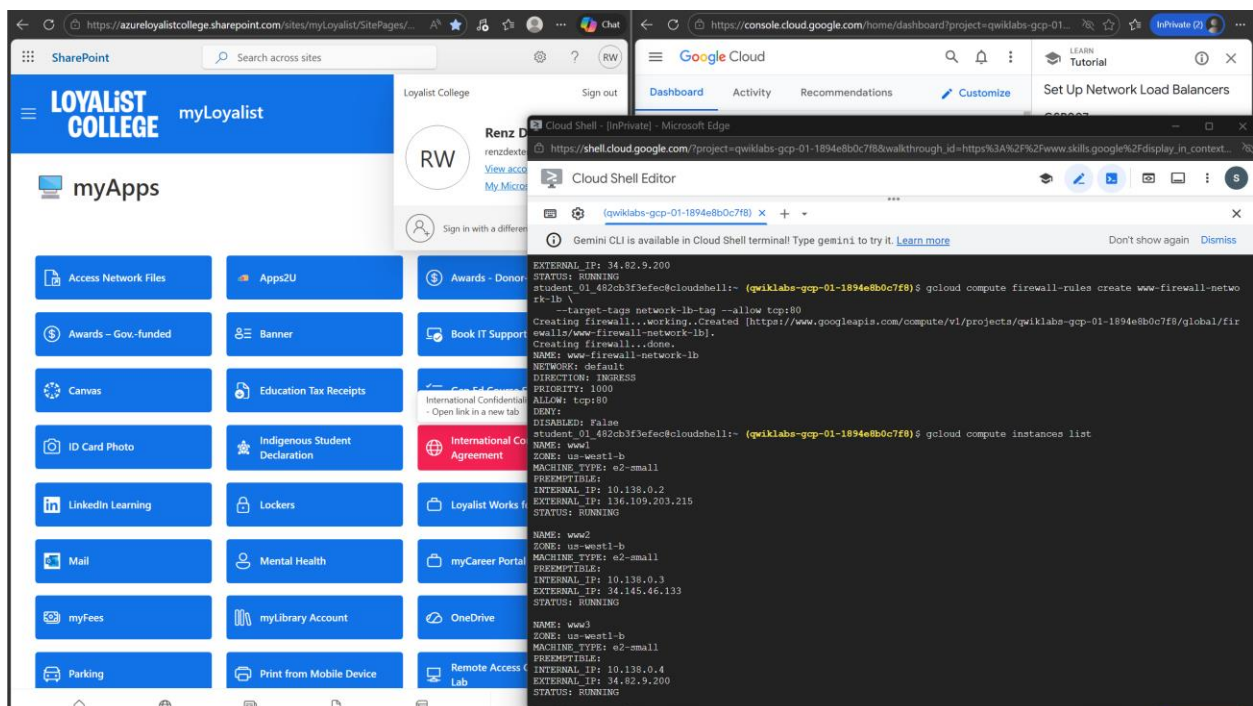


PART 6

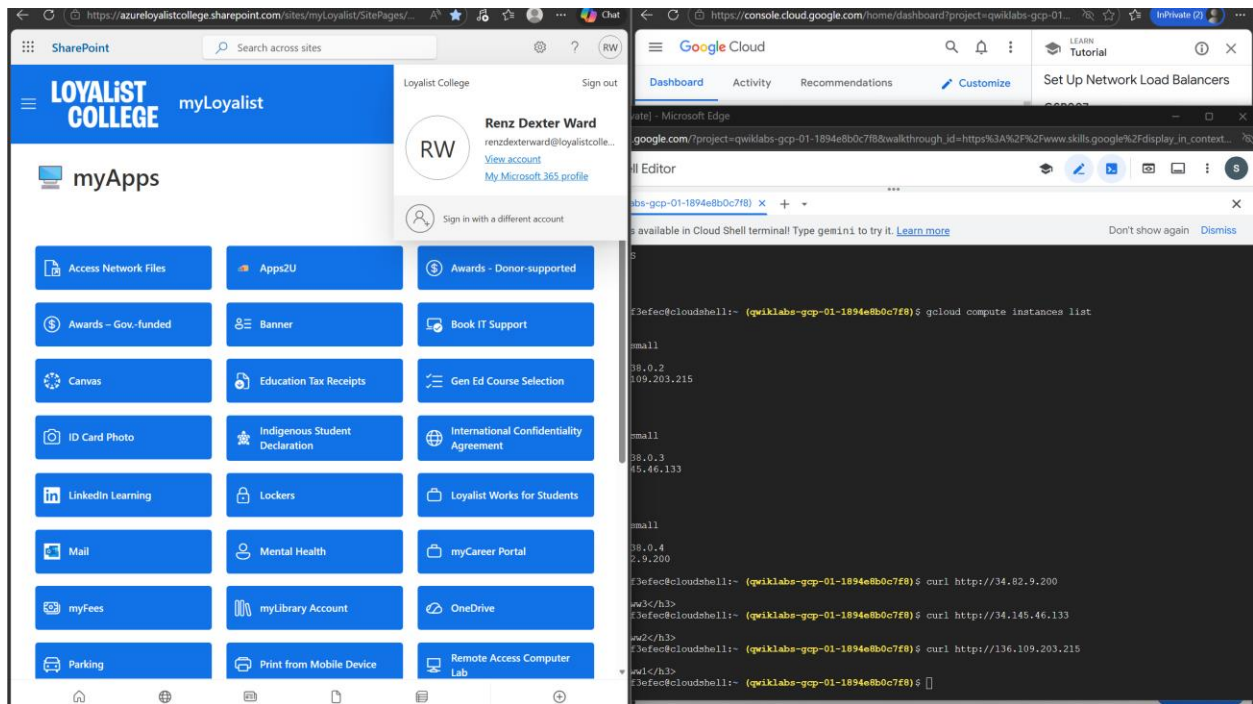
Set the compute region and zone



Created VM Instances and created a firewall rule to allow http traffic into the vm instances

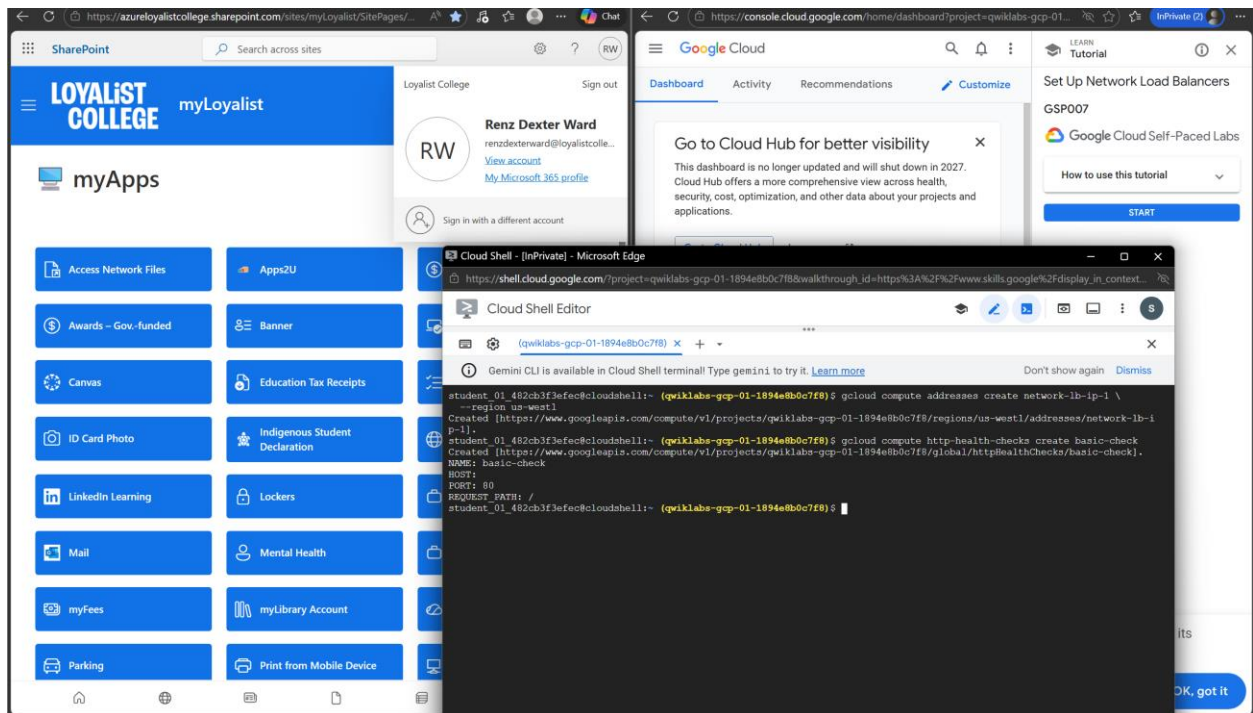


Confirmed the vm instances are running with “curl”



##Configuring the Load Balancers

#Created an external IP for the load balancer. Added a legacy http health check



#Creating the target pool and forwarding rule

The screenshot displays a multi-tab browser interface. The leftmost tab shows the SharePoint 'myLoyalist' portal for 'Loyalist College', featuring a user profile for 'Renz Dexter Ward' and a grid of application tiles such as 'Access Network Files', 'Apps2U', 'Awards - Gov. funded', 'Banner', 'Canvas', 'Education Tax Receipts', 'ID Card Photo', 'Indigenous Student Declaration', 'LinkedIn Learning', 'Lockers', 'Mail', 'Mental Health', 'myFees', 'myLibrary Account', 'Parking', and 'Print from Mobile Device'. The middle tab is the Google Cloud console dashboard for project 'qwklabs-gcp-01-1894e8b0c7f8', showing a 'Go to Cloud Hub for better visibility' message. The rightmost tab is a Cloud Shell terminal window titled 'Cloud Shell - [InPrivate] - Microsoft Edge', which contains the following commands and output:

```
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute addresses create network-lb-ip-1 \
--region us-west1
Created [https://www.googleapis.com/compute/v1/projects/qwklabs-gcp-01-1894e8b0c7f8/regions/us-west1/addresses/network-lb-ip-1].
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute http-health-checks create basic-check \
--region us-west1 --http-health-check basic-check
Created [https://www.googleapis.com/compute/v1/projects/qwklabs-gcp-01-1894e8b0c7f8/global/httpHealthChecks/basic-check].
NAME: basic-check
HOST:
PORT: 80
REQUEST_PATH: /
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute target-pools create www-pool \
--region us-west1 --http-health-check basic-check
Created [https://www.googleapis.com/compute/v1/projects/qwklabs-gcp-01-1894e8b0c7f8/regions/us-west1/targetPools/www-pool].
NAME: www-pool
REGION: us-west1
SESSION_AFFINITY: NONE
BACKUP:
HEALTH_CHECKS: basic-check
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute target-pools add-instances www-pool \
--instances www1,www2,www3
Updated [https://www.googleapis.com/compute/v1/projects/qwklabs-gcp-01-1894e8b0c7f8/regions/us-west1/targetPools/www-pool].
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute forwarding-rules create www-rule \
--region us-west1 \
--ports 80 \
--address network-lb-ip-1 \
--target-pool www-pool
Created [https://www.googleapis.com/compute/v1/projects/qwklabs-gcp-01-1894e8b0c7f8/regions/us-west1/forwardingRules/www-rule].
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $
```

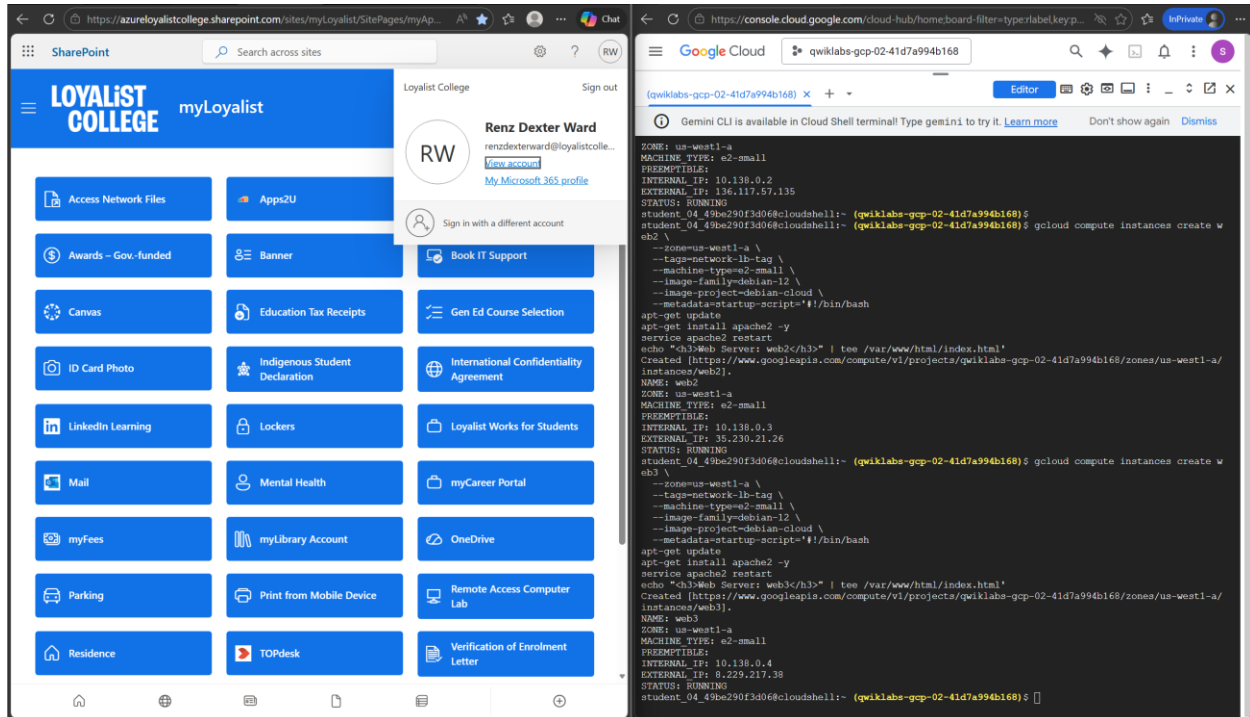
#Sending traffic to the instances

This screenshot shows the same multi-tab browser setup as the previous one. The leftmost tab is the SharePoint 'myLoyalist' portal. The middle tab is the Google Cloud console dashboard. The rightmost tab is a Cloud Shell terminal window showing the following commands and output:

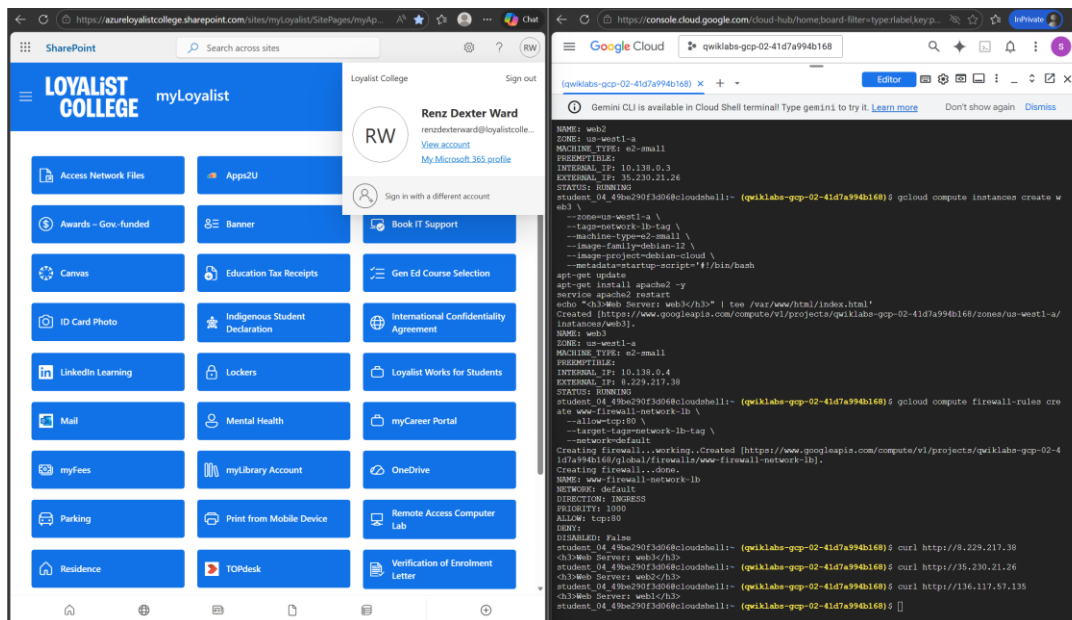
```
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute target-pools add-instances www-pool \
--instances www1,www2,www3
Updated [https://www.googleapis.com/compute/v1/projects/qwklabs-gcp-01-1894e8b0c7f8/regions/us-west1/targetPools/www-pool].
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute forwarding-rules create www-rule \
--region us-west1 \
--ports 80 \
--address network-lb-ip-1 \
--target-pool www-pool
Created [https://www.googleapis.com/compute/v1/projects/qwklabs-gcp-01-1894e8b0c7f8/regions/us-west1/forwardingRules/www-rule].
student_01_482cb3f3fec@cloudshell:~ (qwklabs-gcp-01-1894e8b0c7f8) $ gcloud compute forwarding-rules describe www-rule --region us-west1
2024-03-28T19:00:46.311-07:05
name: www-rule
region: us-west1
ports: [80]
address: network-lb-ip-1
targetPool: www-pool
EXTERNAL
```

Part 7

#Created the web server instances

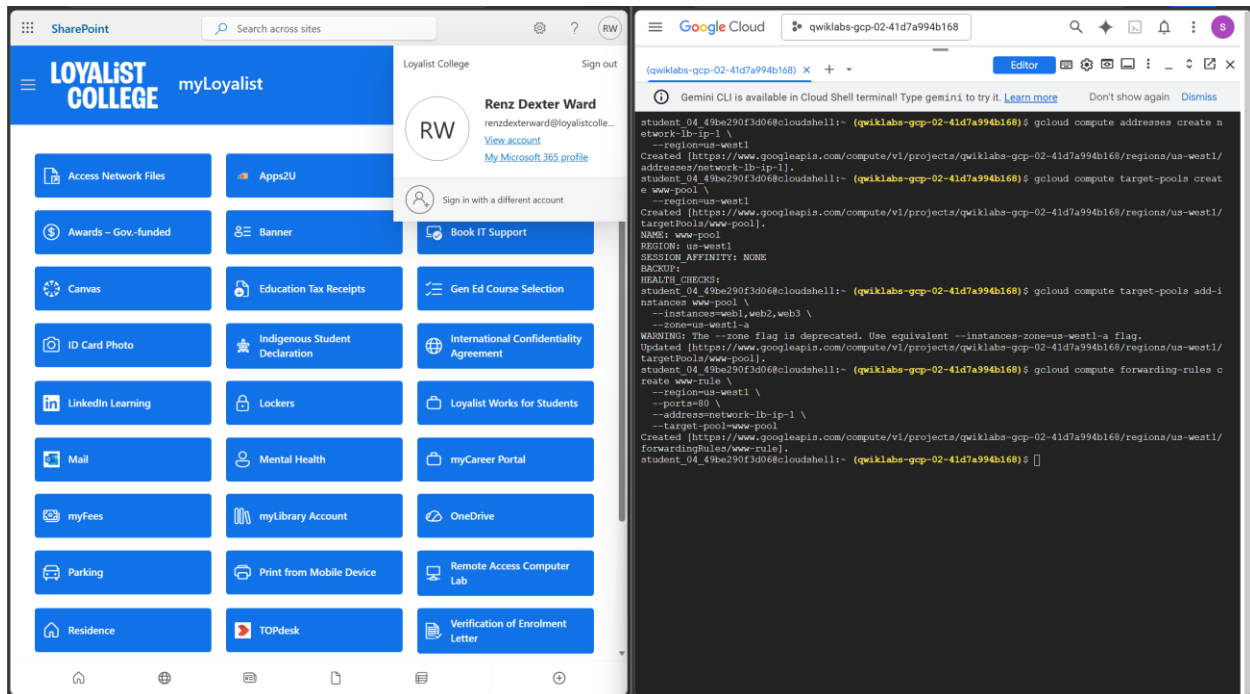


#Created a firewall forward rule and verified the web server instances are running

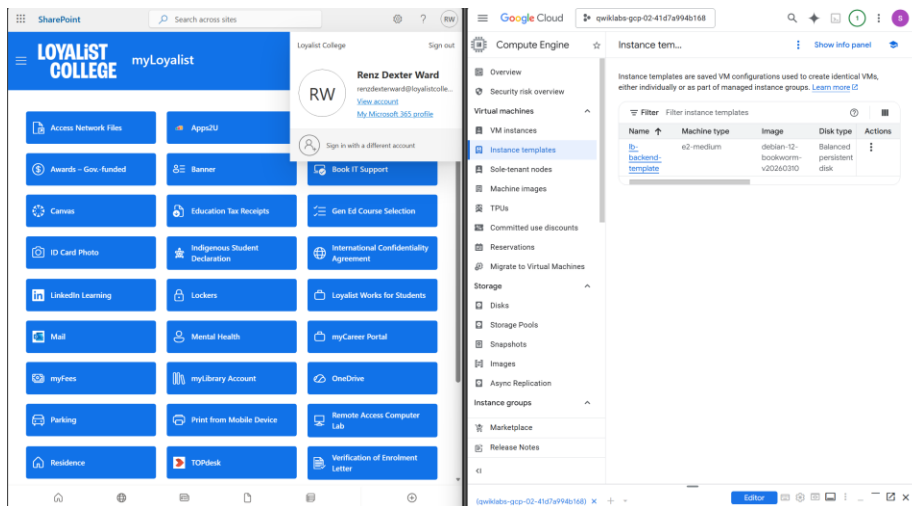


#Configuring the network load balancer

#Reserved the static IP, created target pool, added instances to the pool and created a forwarding rule



#Created instance template



#Created the managed instance group

The image shows two side-by-side browser windows. The left window displays the 'myLoyalist' SharePoint site for Loyalist College, with a user profile for Renz Dexter Ward. The right window shows the Google Cloud console for the 'lb-backend-group' instance group. The console displays the 'Overview' tab, showing the group's status as 'Ready', creation time, and a table of VM instances.

VM instances	Details	Monitoring	Errors
Instances by status	1 instance	Instance by health	Autoscaling
Not configured	Autoscaling off	Configure	Configure

VM instances	Name	Creation Time	Template	Connect
lb-backend-group-2m3c	lb-backend-group-2m3c	Mar 29, 2026, 11:02:31 PM UTC-04:00	lb-backend-template (Regional)	SSH

#Created the health check

The image shows two side-by-side browser windows. The left window displays the 'myLoyalist' SharePoint site for Loyalist College. The right window shows the Google Cloud console for the 'http-basic-check' health check. The console displays the 'Overview' tab, showing the health check's configuration, including the path, protocol, port, and thresholds.

Health checks	Details	Edit	Delete
http-basic-check	In use by	None	
Path	/		
Protocol	HTTP		
Port	80		
PROXY protocol	None		
Logs	Disabled		
Interval	5 seconds		
Timeout	5 seconds		
Healthy threshold	2 consecutive successes		
Unhealthy threshold	2 consecutive failures		
Tags	—		

